Department of Electrical & Electronics Engineering

Abdullah Gül University

Design and Implementation of an Equalizer for Amplitude-Shift Keying (ASK) Modulation using LabVIEW and USRP-2900

Submitted by: Oğuzhan Alasulu

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Team Members:

Oğuzhan Alasulu - 2111011051

Furkan Durmuş - 2011011050

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OBJECTIVE

The aim of this project is to transfer sounds between transmitter and receiver using Amplitude Shift Keying (ASK) modulation and digital signal processing (DSP) techniques. This is done by utilizing the NI LabVIEW software and the NI2900 device. ASK modulation, which forms the basis of the project, carries information using the wave amplitude of audio signals. DSP techniques optimize these signals, minimizing distortions that may occur during transmission and increasing the efficiency of the system while maintaining sound quality. The NI2900 device is used to transmit audio signals and RF coaxial cables with 2 male SMAs are used instead of antennas for transmission. This project demonstrates not only the ability to digitally transmit audio signals, but also the power of DSP and telecommunication techniques in audio processing and the ability of software such as LabVIEW to control these processes. Ultimately, we aim to develop a robust and reliable audio transmission system for different audio signals and transmission media. This project will be very useful for engineers and researchers working in the fields of communication systems and audio processing.

BACKGROUND

In telecommunication technologies, clear transmission of data stands as a fundamental process for effective communication systems. Amplitude Shift Keying (ASK) modulation appears as a common technique for encoding digital data over communication channels by changing the amplitude of a carrier signal. Nevertheless, the employment of ASK modulation includes some challenges such as channel noise, frequency selective fading, and multipath propagation. These challenges can introduce distortions that degrade the quality of transmitted signals, leading to errors in data recovery and transmission.

In that manner, this project aims at the development of an innovative equalization algorithm with advanced Digital Signal Processing (DSP) methodologies to overcome the obstacle of ASK-modulated communication systems. The equalizer will merge 3 different channels with Butterworth filtering and error correction mechanism to efficiently address channel distortions and strengthen the robustness of ASK-modulated signals. The first channel is 20 Hz - 2 kHz which covers bass & midrange intervals. The second channel has an interval of 2 kHz - 8 kHz which consists of upper mid and presence intervals. Finally, the third channel 8 kHz - 20 kHz is named treble (brilliance) interval. Normally, the interval from 2 kHz to 20 kHz is called treble. Additionally, the equalizer will be operated offline utilizing DSP processing techniques to minimize computational delays and verify signal processing.

ASK Modulation facilitates the encoding of digital data by modulating the amplitude of the carrier signal. Complementing that, DSP algorithms are readily adaptable on digital platforms, facially integrating with existing telecommunication systems such as the NI LabVIEW ecosystem. This natural scalability not only provides the adaptability of an equalization solution to improve communication

standards but also emphasizes its longevity and relevance in telecommunication technologies. Therefore, through the unity of ASK modulation and DSP algorithms, this project endeavors to reliability, quality, and robustness of ASK-based communication systems.

DESIGN AND TEST PROCEDURES

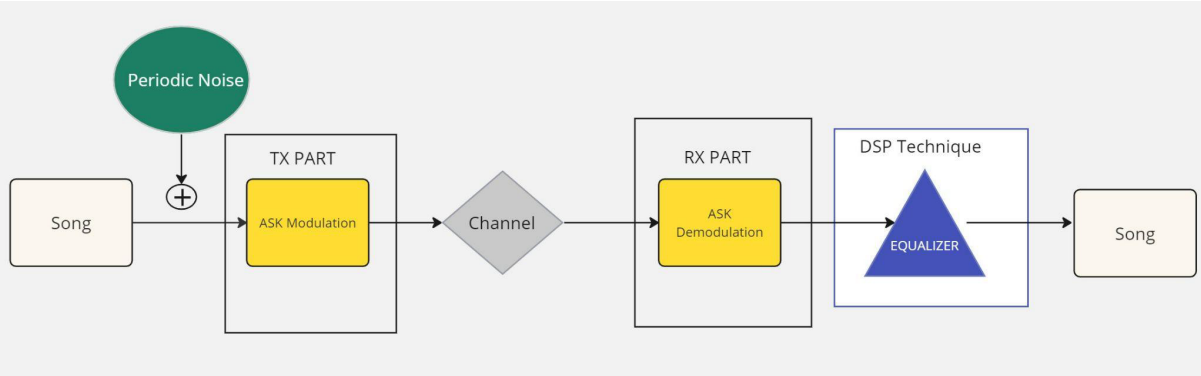

***Fig.* 1** *Flowchart of the Project*

Figure.1 shows the flowchart of the project in simplified view. As shown, the song is taken and periodic noise is added to the input (song) signal. Then, ASK modulation transmits data through the employing the operation of amplitude shift keying with binarizing the input with converting 0 and 1. This part generally is called the transmitter (TX) part. After that, along the channel data has been transmitted and reached to the receiver (RX) part of the system. RX part starts with demodulation. Next, the equalizer part is employed to control the frequency bands and volume of the received song signals. Equalizer part also contributes to lessen the periodic noise and the output data is plates as sound in the end of the system.

Amplitude Shift Keying (ASK) TX Circuit

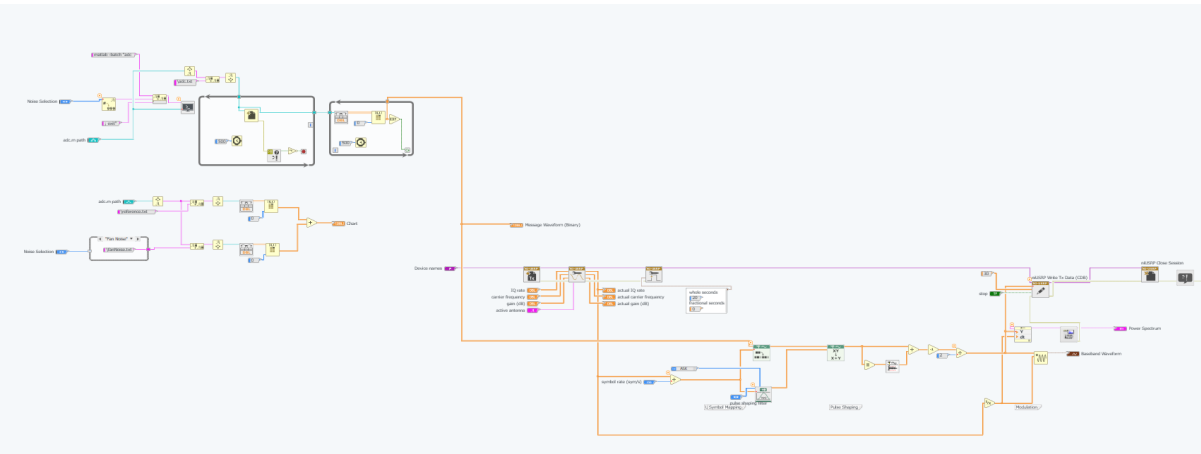

***Fig.* 2** *Transmitter (TX) LabVIEW Diagram*

The transmitter (TX) part of this project is conducted with respect to the ASK TX lab methodology as seen in Figure.2 and Figure 3. Firstly, there is the output which is obtained from ADC song and noise added. For a better understanding of the effect of noise, the panel shows binarized noise. If the bit value is equivalent to 1, it is equivalent to 1 when the bit value 0 corresponds to -1.

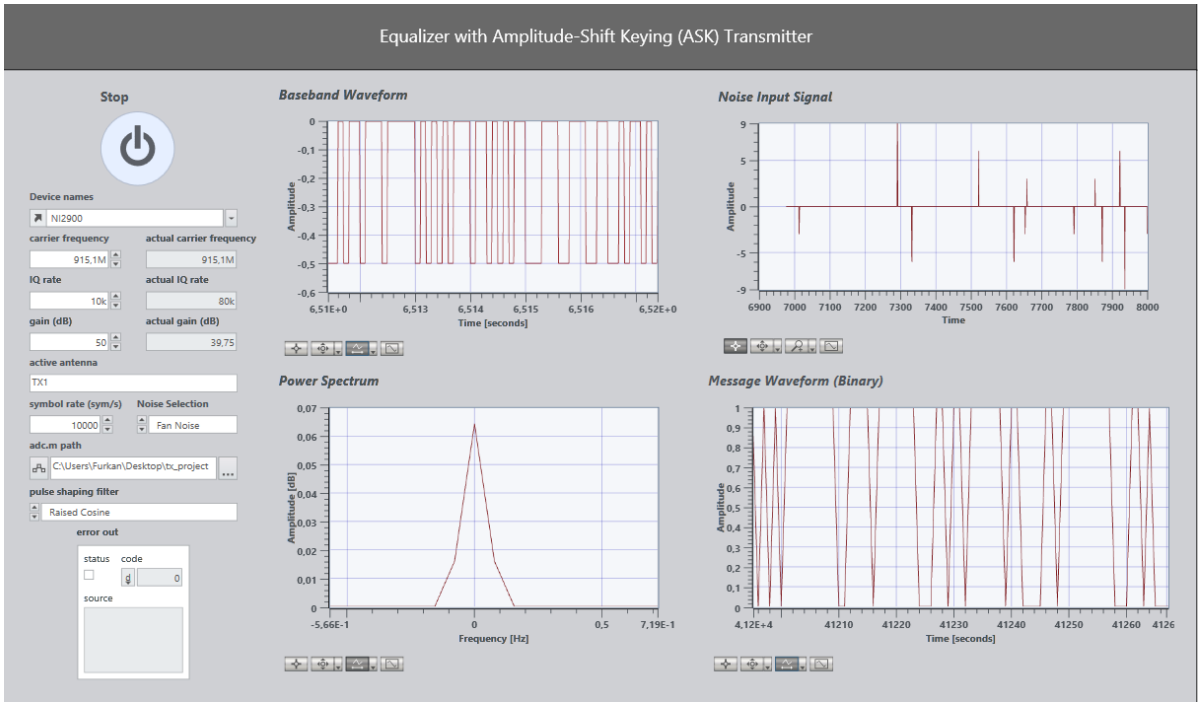

***Fig.* 3** *Transmitter (TX) LabVIEW Panel*

Providing the mathematical expressions of the operations would be benefical for better comprehension about the ASK transmitter system. In that manner, signal processing chain as illustrated in $Fig.$ 2 will be eloborated and explained at below:

<u>Input Bitstream:</u>

$$b[n] \in \{0,1\}$$

<u>Symbol Mapping:</u>

$$s[n] = \begin{cases} A_0 & if\ b[n] = 0 \\ A_1 & if\ b[n] = 1 \end{cases}$$

Where A_0 and A_1 are the amplitude levels corresponding to binary “0” and “1”.

<u>Pulse Shaping Filter:</u>

$$s_{ps}(t) = s[n] * h_{ps}(t)$$

Where $h_{ps}(t)$ is the impulse response of pulse shaping filter

<u>Upsampling, Modulation and, Transmitted Signal:</u>

The upsampling operation is showed at below with equation (1) where T_s is the symbol period, modulation with equation (2) where f_c is the carrier frequency and transmitted signal with equation (3). At the final step, summary of the output will be written in equation (4)

$$s_u(t) = s_{ps}(t) \,.\, \delta(t - nT_s) \quad (1)$$

$$x(t) = s_u(t) \,.\, \cos(2\pi f_c t) \quad (2)$$

$$x(t) \quad (3)$$

$$x(t) = \left(\left(\sum_n s[n] * h_{ps}(t) \right) . \delta(t - nT_s) \right) . \cos(2\pi f_c t) \quad (4)$$

The mathematical expression of the ASK transmitter scheme consists of a series of calculations to convert the input bitstream into a modulated signal ready for transmission as written in (4). This equation takes the digital signal processing chain for ASK TX system, involving symbol mapping, pulse shaping, upsampling and carrier modulation.

ADC

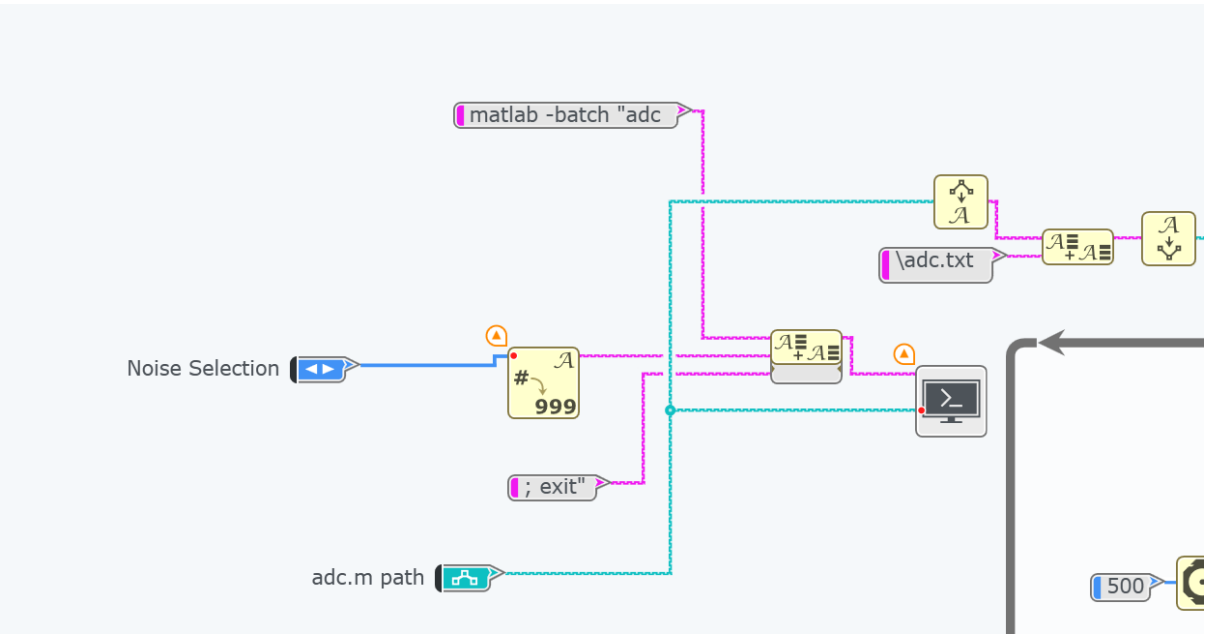

Fig.4 *Analog − to − Digital Conversion on LabVIEW with MATLAB*

LabVIEW program utilizes an external file (ADC.m) to process data using an Analog-to-Digital Converter (ADC) as seen in Figure 4. Reyes et. al (2017) demonstrates that through a special ADC system, the bit-error rate can be reduced to fall below the 0.1 dB from the 1dB. The user selects the location of this file through a graphical interface element (panel). Before generating a new output file (adc.txt) containing the converted data stream (bitstream), any existing file with the same name is deleted to prevent conflicts. A loop continuously checks for the creation of the new output file (adc.txt). Once the file is detected, the loop terminates, and the file path is passed to a separate program component responsible for reading data from spreadsheets (Read from Spreadsheet File VI).

The "adc.m" MATLAB code employs numerous steps to convert an analog signal to digital representation. It basically comprises of combining signals, quantization the levels, quantization, binary conversion and output reshaping. Briefly, all these processes will be described below respectively:

Combining Signals:

The reference signal $r(t)$ and particular noise signal $n(t)$ are summed where $d(t)$ is the combined information signal:

$$d(t) = r(t) + n(t)$$

Quantization of the Levels and Quantization:

Defining the quantization levels with using 16 bits which creates 65536 equally spaced quantization levels between -1 and 1

$$partition[i] = -1 + \frac{2i}{2^{-16} - 1} \quad i = 0, 1, 2, \dots, 2^{16} - 1$$

The information signal $d(t)$ is quantized, the operation at below finds the index of the closest quantization level for each sample in $d(t)$.

$$index(t) = \arg\min |d(t) - partition(i)|$$

Binary Conversion and Output Reshaping:

The binary representation is then converted to an array of individual bits where each index is converted to 16-bit binary number:

$$binary[i] = dec2bin(index|i], 16)$$

The output will be converted to an array of the individual bits.

Synchronization

When the synchronization part is transferred from the transmitter side to the receiver side, there is a high delay, and the data is not completely transferred to the receiver side. Also, the second problem is that the samples are not completely transferred to the receiver part and the receiver part acts as a buffer. Due to this, the "NI USRP configure section, the time delay has been stabilized to 20 seconds in accordance with the system and the high delay amount (around 1 minute) when transferring the audio file to the transceiver section has been reduced by 10 seconds as depicted in Fig.5

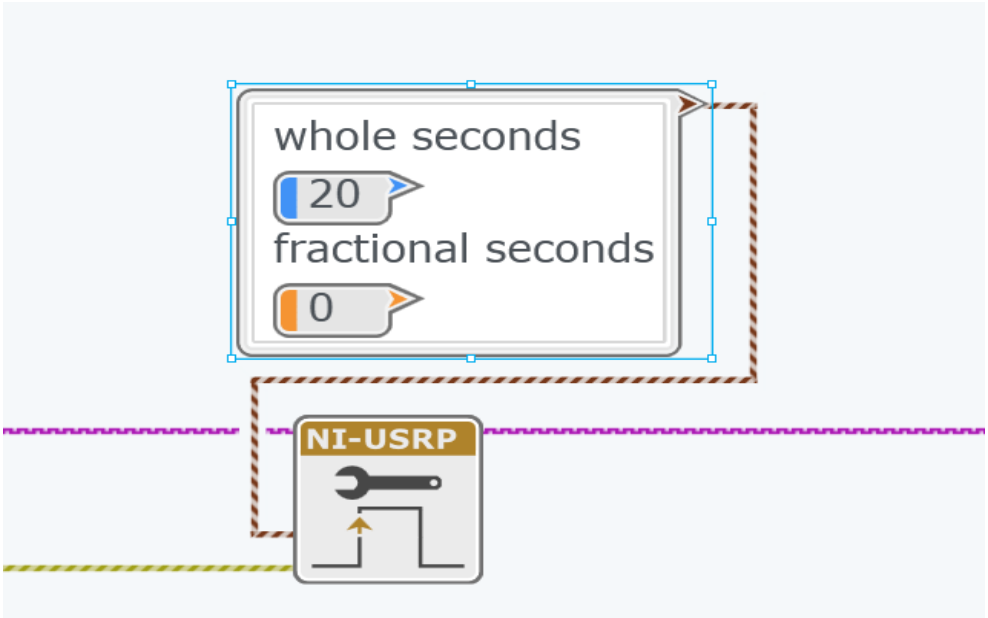

***Fig.* 5** *Synchronizer Schematic for Transmitter Diagram*

Amplitude Shift Keying (ASK) RX Circuit

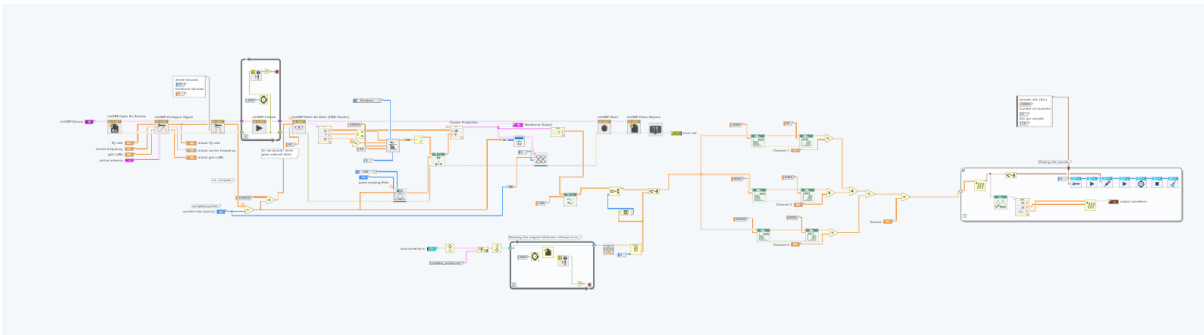

***Fig.* 6** *Receiver (RX) LabVIEW Diagram*

First, the number of samples in a frame is estimated by considering the message length, symbol rate and the actual IQ of the system. This information is then used to link the frame's output to the demodulation stage. ASK demodulation typically involves filtering techniques like FIR filters or

median filters to remove noise and improve signal quality. The demodulated signal is then analyzed using an eye diagram to assess signal integrity. Finally, the recovered digital bits can be visualized alongside the demodulated signal for clarity. An equalizer block is connected to the last part of the ASK block as shown in Figure 6. Also, at the end of this block, the signal is converted to digital signal using the DAC block and the system is completed using the playing sound block.

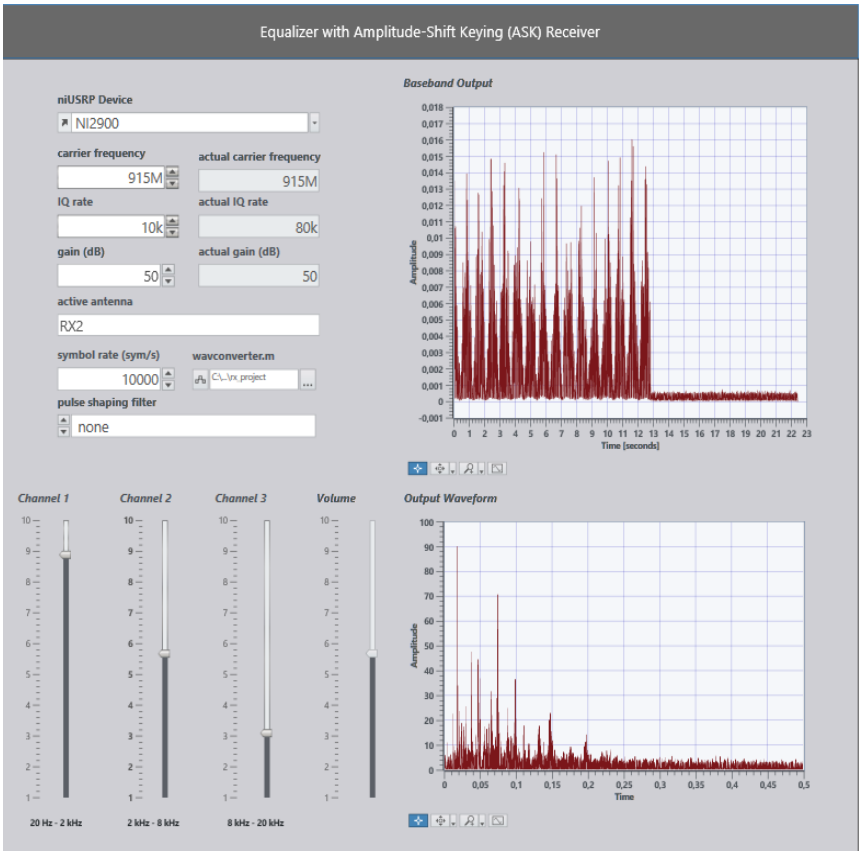

Fig.7 *Receiver (RX) LabVIEW Panel*

Containing the mathematical expressions for the processes can improve understanding of the ASK receiver system. Thus, the signal processing chain depicted in Fig.6 will be explained and enlightened below, with a focus on these mathematical details. Operation processes consist of with receiving signal, bandpass filtering, mixing/demodulation, pulse shaping filtering, downsampling, matched filtering and, decision making.

<u>Receiving Signal and Bandpass Filtering:</u>

$$r(t) = s(t) \,.\cos(2\pi f_c t) + n(t) \qquad (1)$$

$$r_{bp}(t) = r(t) * h_{bp}(t) \qquad (2)$$

Where $s(t)$ is the transmitted ASK signal, f_c is the carrier frequency and $n(t)$ in the noise in equation (1) and $h_{bp}(t)$ is the impulse response of the bandpass filter in equation (2).

Demodulation and Pulse Shaping Filtering:

$$r_{bb}(t) = r_{bp}(t) \,.\cos(2\pi f_c t) \quad (3)$$
$$r_{ps}(t) = r_{bb}(t) * h_{ps}(t) \quad (4)$$

The equation (3) shifts the signal to the baseband. In the equation (4), $h_{ps}(t)$ is the impulse response of the pulse shaping filter.

Downsampling and Matching Filtering:

$$r_d[n] = r_{ps}(nT_s) \quad (5)$$
$$r_m[n] = r_d[n] * h_m[n] \quad (6)$$

In the equation (5), T_s is the symbol period. In the equation (6), $h_m[n]$ is the impulse response of matched filter.

Decision Making:

$$\hat{b} = \begin{cases} 1 \; if \; r_m[n] \geq r \\ 0 \; if \; r_m[n] < r \end{cases} \quad (7)$$

DAC

A Digital to Analog Converter (DAC) is a method used to convert digital signals into analog signals. DACs represent digital signals using a specific number of bits. The most common DACs have 8-bit, 16-bit, 24-bit and 32-bit resolutions. The system designed in this project is 16 bits. MATLAB code was written for the DAC using a mathscript node. In this code, each 16-bit data is converted back into decimal forms. These values are the division values of the data. Therefore, to find the point values of these values, the code finds the corresponding values in the partition array and stores them in matrices. At the output, we have the floating-point values of the corrected signal. The operation is shown in Fig.8

```
partition = linspace(-1, 1, pow2(16));

b2d = zeros(8000,1);
errb2d = zeros(8000,1);
for i = 1 : length(decimalToBinary)
    b2d(i, :) = sum(decimalToBinary(i, :) .* pow2(15:-1:0))
    errb2d(i, :) = sum(errored(i, :) .* pow2(15:-1:0));
end

b2d(b2d == 0) = 1;
errb2d(errb2d == 0) = 1;

digitalAnalog = [];
erroredSignal = [];
for i = 1 : length(b2d)
    digitalAnalog = [digitalAnalog; partition(b2d(i))];
    erroredSignal = [erroredSignal; partition(errb2d(i))];
end
```

Fig.8 *Receiver (RX) LabVIEW Panel*

DSP Part

The DSP component of our LabVIEW project focuses on the design and application of a three-channel equalizer as appeared in Fig.9, covering specific frequency ranges to improve audio quality with precise frequency band control and volume adjustment. Shokri et. al (2017) states that equalizers provide better voice quality and robustness against the channel noise

The first channel of our equalizer consists of the frequency interval from 20 Hz to 2 kHz, containing the bass and midrange frequencies. The exact bass frequency range (20 Hz- 250 Hz) is separating two branches. First branch is a sub-bass which is located between 20 Hz-60 Hz, providing deep rumble and low-end richness. Second branch has the range from 60 Hz to 250 Hz and is called a bass which adds warmth and body to the sound. The midrange frequency interval is issued on the frequency interval of 250 Hz - 2 kHz and that is crucial for clarity and intelligibility in both music and speech.

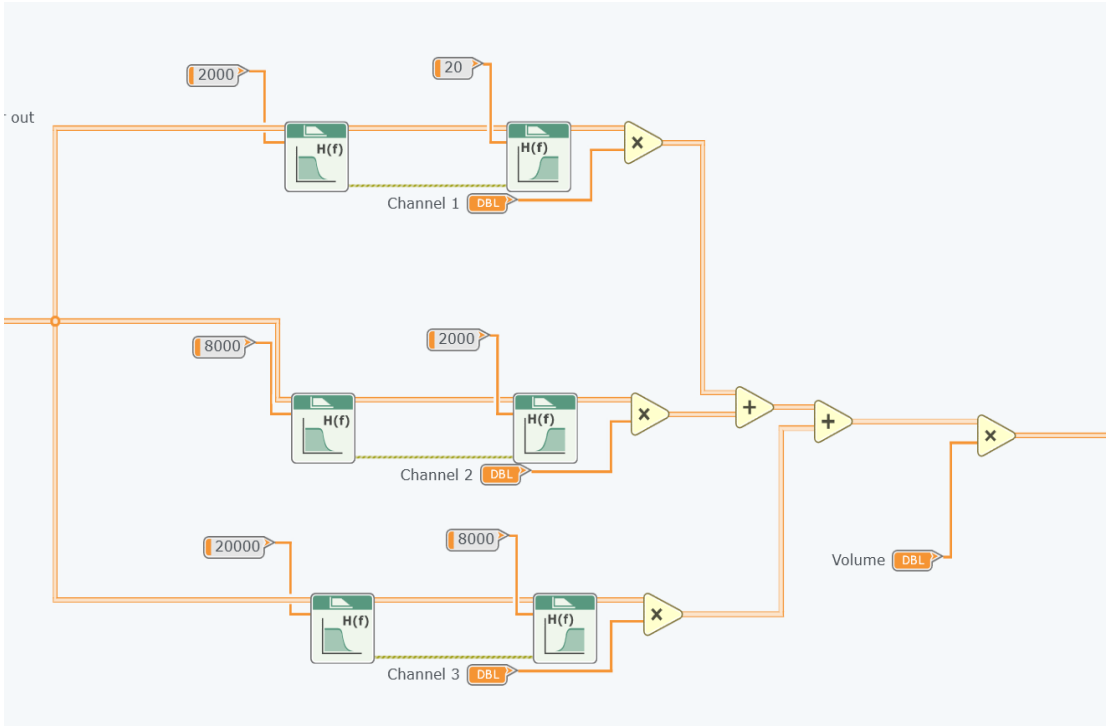

***Fig.* 9** 3 *Channel Equalizer in Receiver Diagram*

The second channel spans from 2 kHz to 8 kHz, including the upper-midrange and presence frequencies. The upper midrange is located between 2 kHz and 5 kHz which contributes to the presence, clarity and detail. Presence frequencies have a range from 5 kHz to 8 kHz to enhance clarity, adding sharpness and proximity to the audio, essential for high-fidelity sound imitation.

The third channel provides brilliance frequencies from 8 kHz to 20 kHz, adding airiness to the sound, improving detail and texture with a sense of space and openness. The broader range from 2 kHz

to 20 kHz is called a treble which contributes to clarity, brightness and definition with clean audio experience.

The diagram of equalizer operates Butterworth Sallen-Key filters for low-pass and high-pass filters for each frequency interval. Butterworth filters are named for their maximally straight frequency response in the passband, providing smooth amplification or attenuation without ripples. In each channel of the equalizer is implemented with using Butterworth Sallen-Key filter specific to frequency intervals for a smooth frequency response for every channel and concluding in clear and high-quality audio output. The equalizer admits users to enhance or attenuate specific frequency ranges respect to their choices, with a volume slider for extra control over the overall output level.

Playing Audio Circuit

In order to listen to the sound of the transmitted signal, the sound block was used and inserted into the for loop and the sound sampling was set to 8000 for optimal listening. As can be seen in Figure 10, the sampling frequency was set to 8000 and the bits per sample was set to 16 bits/s. As a result, the signal is successfully modulated, demodulated and converted to its previous form.

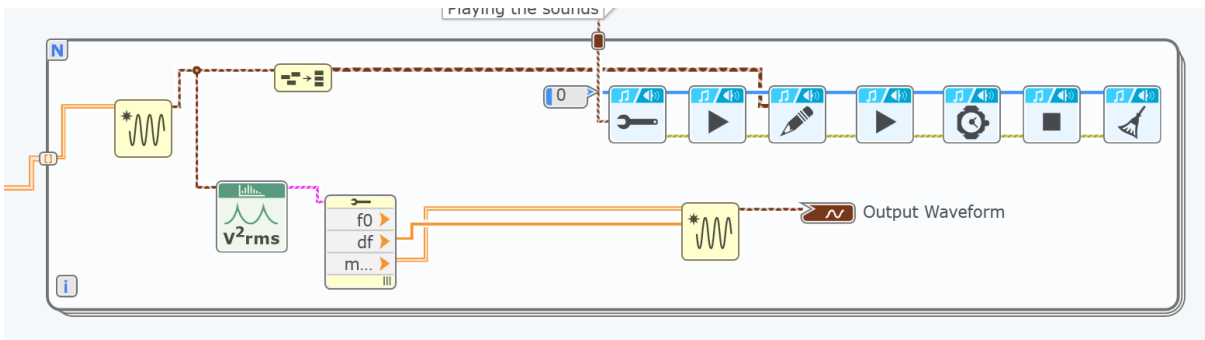

***Fig.* 10** *Voice Playing Diagram in Receiver (RX)*

CONCLUSION

In this project, we aim to transmit messages over a channel using Amplitude Shift Keying (ASK) modulation and demodulation techniques. The signal is transmitted from the TX side to the RX side on the NI LABVIEW program using both an adaptive equalizer algorithm and signal processing stages by converting the signals into digital and analog media with Analog to Digital Converter (ADC) and Digital to Analog Converter (DAC). An adaptive equalization algorithm was used to eliminate distortions in the signals and provide clearer communication. We also investigated the role of ADC and DAC in the conversion of signals. In this way, the functioning of message communication systems, signal processing techniques and equalization algorithms were used. As a result of this project, we aim to gain valuable knowledge and experience for anyone interested in message communication systems and signal processing techniques.

References

[1] Reyes, B. T., Sanchez, R. M., Pola, A. L., & Hueda, M. R. (2016). Design and experimental evaluation of a time-interleaved ADC calibration algorithm for application in high-speed communication systems. *IEEE Transactions on Circuits and Systems I: Regular Papers*, *64*(5), 1019-1030.

[2] Shokri, S., Ismail, M., Zainal, N., & Moghaddasi, M. (2017). Audio-speech watermarking using a channel equalizer. *Wireless Personal Communications*, *95*, 4457-4476.